



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.

Dataset at a Glance

3,900

Purchases

Total transactions across all categories
Demographics, purchase details
& behavior

18

Features

Purchase & Behavior

Item, Category, Amount, Discount, Shipping,
Review Rating, Frequency

37

Missing Values

In Review Rating column only

Customer Demographics

Age, Gender, Location, Subscription Status

Data Preparation in Python

01

Load & Explore

Imported dataset via pandas; used `df.info()` and `.describe()` for structure and summary

02

Clean & Standardize

Imputed missing Review Ratings using median by category; renamed columns to snake

03

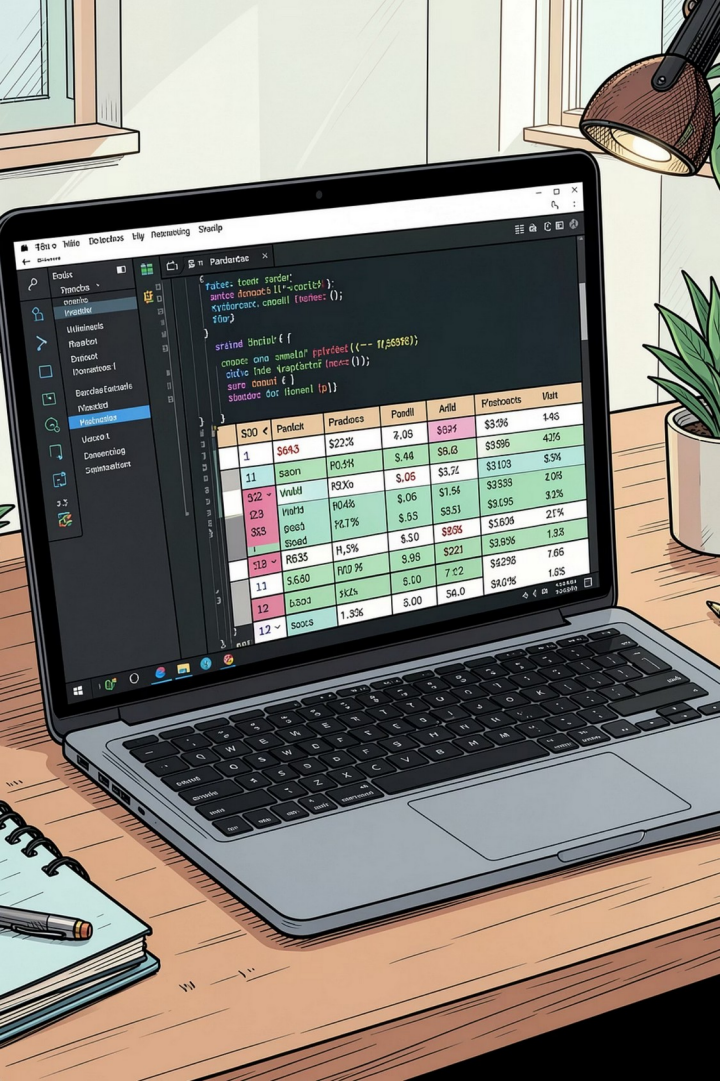
Feature Engineering

Created `age_group` and `purchase_frequency_days`; dropped redundant `promo_code_used`.

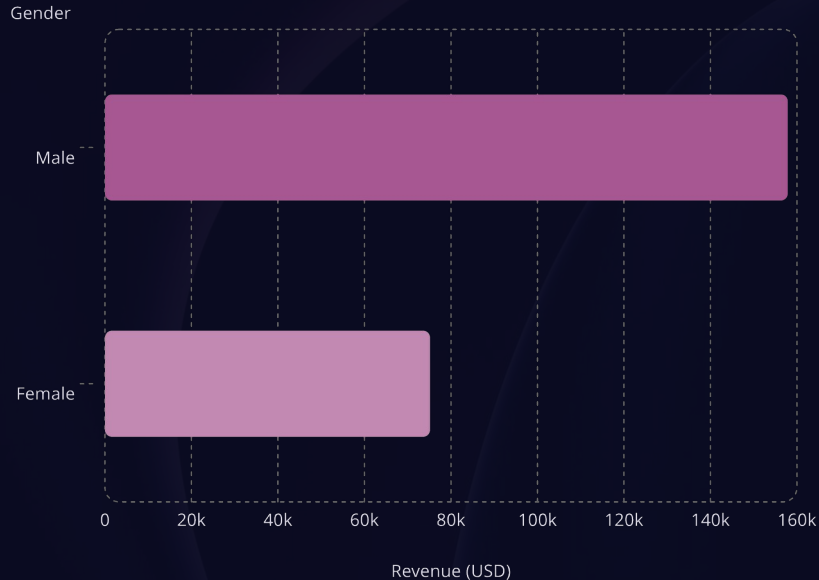
04

Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis.



Revenue by Gender



Key Insight

Male customers generated **\$157,890** in revenue — more than double the \$75,191 from female customers.

Male shoppers accounted for 2,652 of 3,900 transactions, making them the dominant purchasing segment.

Top Products & Shipping Insights

Top 5 Products by Rating



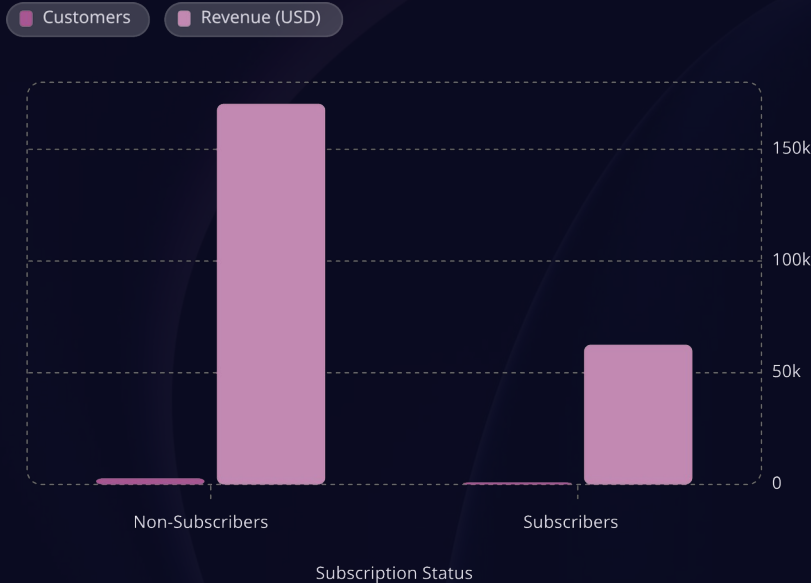
Shipping Type Comparison

Express shipping averaged **\$60.48** vs. **\$58.46** for Standard — a modest \$2.02 premium per order.

Discount-Dependent Products

Hats, Sneakers, and Coats had the highest discount rates (~49–50%), suggesting strong price sensitivity in these categories.

Subscribers vs. Non-Subscribers



Key Findings

Non-subscribers dominate with **2,847 customers** and **\$170,436** in total revenue. Subscribers (1,053) contributed \$62,645.

Average spend is nearly identical: **\$59.87** (non-subscribers) vs. **\$59.49** (subscribers), indicating subscription status doesn't significantly affect per-order value.

Customer Segmentation

🏆 Loyal

3,116 customers — the vast majority of the base

↺ Returning

701 customers — repeat buyers with growth potential

NEW New

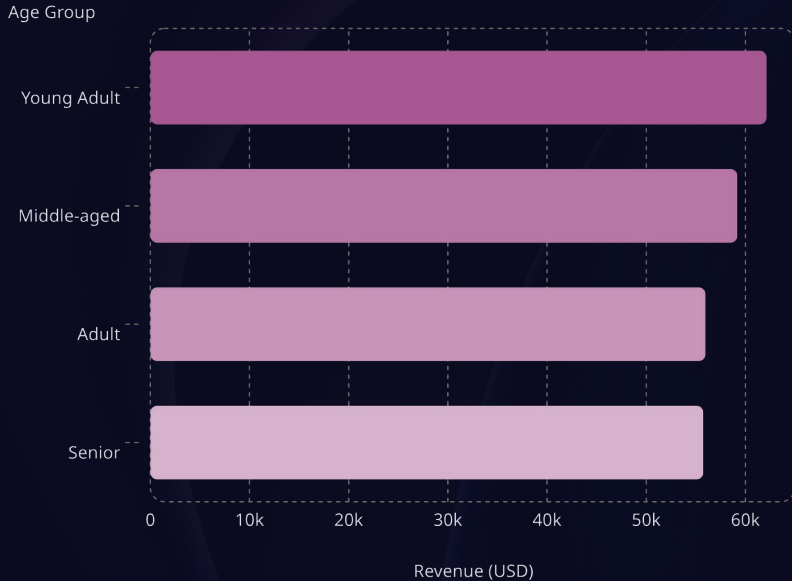
83 customers — first-time buyers, smallest segment

Of customers with more than 5 purchases, **958 are subscribers** vs. 2,518 non-subscribers — repeat buyers show higher subscription uptake.



Revenue by Age Group & Top Products

Revenue by Age Group



Top Products per Category

- **Accessories:** Jewelry (171), Sunglasses (161), Belt (161)
- **Clothing:** Blouse (171), Pants (171), Shirt (169)
- **Footwear:** Sandals (160), Shoes (150), Sneakers (145)
- **Outerwear:** Jacket (163), Coat (161)

Young Adults lead revenue at **\$62,143**, but all age groups contribute relatively evenly.

Power BI Dashboard



Dashboard Highlights

- **4K Customers** · **\$59.76** Avg Purchase · **3.75** Avg Rating
- **92.7%** non-subscribers vs. 7.3% subscribers (donut chart)
- Revenue & sales breakdowns by category, age group, and shipping type
- Interactive filters: subscription status, gender, category, shipping type

Business Recommendations

Boost Subscriptions

Promote exclusive benefits to convert the 92.7% non-subscriber base.

Customer Loyalty Programs

Reward repeat buyers to grow the Loyal segment beyond 3,116 customers.

Review Discount Policy

Balance sales boosts with margin control — ~50% discount rates on Hats and Sneakers may erode profitability.

Targeted Marketing

Focus on Young Adults (top revenue group) and express-shipping users; highlight top-rated products like Gloves and Sandals.

